

# Co-Creating Virtual Experiences: Fostering Inclusivity and Empathy Building

Jinsil Hwaryoung Seo

College of Performance, Visualization & Fine Arts,  
Texas A&M University  
United States of America  
hwaryoung@tamu.edu

Caleb Kicklighter

College of Performance, Visualization & Fine Arts,  
Texas A&M University  
United States of America  
kicklica@tamu.edu

Brittany Garcia-Pi

Institute of Data Science,  
Texas A&M University  
United States of America  
brinni@tamu.edu

Paul Taele

College of Engineering,  
Texas A&M University  
USA  
ptaele@tamu.edu

## Abstract

Virtual reality (VR) can provide transformative educational experiences for individuals with intellectual disabilities (ID). To leverage this aspect in higher education, a team of faculty members and a transitional university program collaborated to integrate VR into their support systems. The initial VR experience sessions led to the development of specialized VR applications designed to support individuals with ID and raise awareness about their needs in the university setting. We detail a co-design process resulting in two VR projects: one for enhancing job interview skills and another for building empathy by simulating challenges faced by individuals with ID. This program highlights the effectiveness of inclusive participatory design in VR development, demonstrating how cross-ability collaboration leads to more accessible and relevant technological solutions. Our findings underscore VR's potential for skill development and fostering empathy, while contributing to the broader discourse on inclusive technology development and promoting a more inclusive society.

## CCS Concepts

• **Index Terms:** Human-centered computing; • **Human computer interaction (HCI);** • **Interaction paradigms;** • **Virtual reality;** • **Human-centered computing;** • **Interaction design;** • **Interaction design process and methods;** • **Participatory design;**

## Keywords

Inclusive Design, Virtual Reality, Collaborative Design, Collaboration, Intellectual disability

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## 1 Introduction

Virtual Reality (VR) has emerged as a transformative technology with far-reaching applications across education, training, entertainment, and therapeutic interventions. As VR continues to evolve, the imperative for inclusivity and diversity in its design and development education becomes increasingly apparent. This paper examines the crucial role of individuals with intellectual disabilities (ID) in VR application creation, focusing on collaborative projects conducted at our university. The traditional barriers faced by individuals with ID in accessing and benefiting from technological advancements underscore the urgency of their inclusion in VR education. This inclusion is not only a matter of equity and ethics but also vital for introducing unique perspectives that can profoundly enrich and guide the trajectory of VR design and innovation.

At the core of our research is the co-design methodology, a participatory approach that facilitates collaboration between expert designers and developers, non-experts, including individuals with ID, throughout the creative process. This methodology, as outlined by Sanders and Stappers, 2008, and further explored in the urban design context by Wates et al. (2014) and Žlender et al. (2021), emphasizes active participation in all stages of the design process. It ranges from initiation to maintenance, ensuring that all stakeholders, including users with ID, actively contribute to and shape the final outcome. This inclusive approach not only negotiates the project's value creation but also fosters a common understanding and mitigates uncertainties arising from information asymmetry, as noted by Liu et al. (2019) and Williams et al. (2019).

Our paper presents our program's process and outcomes, offering valuable insights into the educational landscape and emphasizing the collaborative efforts with individuals with ID. We detail the development of two VR projects: one designed to facilitate job interview practice and another aimed at fostering empathy towards individuals with ID. By sharing our experiences, challenges, and lessons learned, we aim to advocate for a more inclusive approach within the broader domains of VR design and development in higher education.

## 2 VR for People with Intellectual Disabilities (ID)

VR technology offers significant promise in teaching independent living skills to adolescents and young adults with ID (Standen and Brown, 2006). The interactive and immersive nature of VR facilitates the transfer of learned skills from virtual to real-world environments, enabling the mastery of practical skills needed for independent living. In addition, VR excels in enhancing emotionally relevant experiences and fostering social interactions, benefiting both typically developing children and those with disabilities [Parsons et al., 2017]. By simulating real-life scenarios, VR creates a safe and controlled setting for practicing social skills and managing emotional responses, essential for holistic development. Beyond this, VR acts as a valuable supplementary tool to traditional educational services, uniquely increasing skills and motivation among individuals with intellectual and developmental disabilities [Panerai et al., 2018]. Furthermore, its application in cognitive and social skill improvement is noteworthy. VR not only promotes essential skills for independent living but also significantly enhances cognitive performance and social capabilities, underlining its vast rehabilitative potential for people with ID [Oakes, 2022; Standen and Brown, 2005].

While VR offers innovative opportunities for individuals with ID, several challenges need to be addressed for its effective integration [Rizzo et al., 1998]: 1) Cognitive demands-The effectiveness of VR relies on the user's cognitive abilities. Navigating and interacting within a virtual environment can be challenging for those with ID, as they may struggle with understanding, problem-solving, and memory, limiting their full engagement with VR experiences; 2) Cost and accessibility-The high expense of VR technology can restrict its availability in educational and therapeutic settings, where it could significantly benefit young adults with ID; 3) Supervision and support-Safe and effective use of VR often demands supervision and assistance. This includes help with using the equipment, interpreting the VR experience, and managing emotional or behavioral responses; 4) Inclusion in development: People with ID should be integral to the VR development process. While they show interest in creating VR applications, they often have limited capabilities or access. This needs to be carefully considered and addressed in the design and development phases [Standen and Brown, 2005].

To successfully integrate VR into educational programs for individuals with ID, it is crucial to address these challenges by developing intuitive interfaces, increasing accessibility through cost-effective solutions, providing adequate support, and involving individuals with ID in the design process.

## 3 Inclusive Participatory Design

To address the limitations in VR education, our team has adopted participatory design methods, focusing on co-design with young adults with ID. This approach is highly effective for creating inclusive and accessible products, services, or systems by involving end-users as active co-designers. This method is particularly beneficial in ensuring that the needs and preferences of individuals with ID are adequately met [Schuler and Namioka, 1993].

Studies have explored the use of participatory design with individuals with ID. Gonzalez et al. (2020) identified four phases

of the co-design process in their study with adults with ID and stressed the importance of fully incorporating this community into user-centered design processes. Abascal et al., 2020 found that participatory design with people with cognitive disabilities is possible and increases users' affinity to the resulting application. Similarly Malinverni et al., 2014 discovered the positive impact of involving children with special needs in the design process. These studies underscore the effectiveness of involving individuals with ID in creating user-friendly and accessible solutions. However, successful implementation requires appropriate communication methods and assistive technology tailored to the specific needs of participants with ID.

By adopting this inclusive approach, we can ensure that VR technology not only benefits individuals with ID but also reflects their unique perspectives and needs, maximizing its potential as an educational and developmental tool. Moreover, this collaborative process offers significant advantages for neurotypical college students as well. These students gain invaluable hands-on experience in inclusive design practices, enhancing their understanding of accessibility issues and user-centered design principles. This exposure cultivates empathy, broadens their perspective on diverse user needs, and equips them with practical skills highly valued in the technology and design industries. Furthermore, the interdisciplinary nature of VR development allows students from various fields such as computer science, psychology, education, and design to collaborate, mirroring real-world professional environments. This cross-pollination of ideas and skills not only enriches the VR applications being developed but also prepares students for future careers in inclusive technology design.

By bringing together diverse perspectives and abilities, this approach sets a new standard for inclusive technology development, demonstrating how academic institutions can play a pivotal role in creating more accessible and empowering digital futures.

## 4 Program Overview

Our team collaborated with a dedicated group of students and staff from our university, all involved in a comprehensive transition program for young adults with ID. Together, through inclusive participatory design, we worked to create a range of VR applications. The interdisciplinary research team was comprised of undergraduate and graduate students from multiple disciplines including Visualization, Computer Science, Architecture and Psychology. There were two groups, each led by a graduate student who guided a team of undergraduate students through the design and development process. Oversight was provided by a group of faculty advisors. Team leads engaged in discussions with faculty members to explore project details. It was decided that one team would focus on developing an immersive conversational experience (i.e., *AllyChat*) while the other would craft a narrative story aimed at fostering empathy for the experiences of individuals with disabilities (i.e., *Through the Eyes*). In order to create appropriate applications for this special population teams had to work closely with students from the transition program.

## 4.1 Participants

Initially, five students from the transition program experienced VR applications but two male students decided to participate in the design and development process. They were in their early 20's. A program director and a coordinator from the transition program participated in the collaboration to provide health-related and educational support. The *Through the Eyes* team included a master's student as project manager and an undergraduate 3D designer, both from the Visualization Department, along with an undergraduate computer science VR developer. The *AllyChat* team composed of a project manager, a PhD student from the Architecture Department, a master's student developer specializing in conversational AI from computer science, an undergraduate VR developer from the computer science department, and an undergraduate 3D designer from the Visualization Department.

## 4.2 Methodology

To foster a meaningful relationship between the research teams and the transition program students, a designated weekly gathering was established. During this time the student teams dedicated time to interact with the program students. Initially, the gatherings involved introducing the students to VR, as neither of them had experienced it before. Once a relationship was established each research team would meet separately with one of the students to discuss their respective projects, teams alternated which student they spoke with to obtain different perspectives. The section below details the individual process each team went through.

## 5 VR Projects

### 5.1 Through the Eyes: Immersive Narrative

*Through the Eyes* is a 3D-animated short film in VR, that tells the story of a young boy struggling to express himself through writing. The story presents the challenges of dysgraphia, a neurological disorder and learning disability that impairs writing ability [Gary et al., 2023]. This disorder affects fine motor skills and the coherent translation of thoughts into written words. This story was created from multiple conversational sessions with one of the program students. The student shared his memories with the project design students first and the design students sketched their ideas with the program students. Through an iterative process a final story emerged. The main character's struggle is visualized through a stylistic metaphor: floating, emissive symbols hover above the protagonist, symbolizing his untranscribed thoughts. As the boy attempts to write, these symbols transform from legible letters into gibberish and illegible script, mirroring his frustration. The symbols then morph into pictograms of animals that elude the boy, who must chase and recapture these words to convert them back into his written thoughts.

This narrative unfolds in a 360-degree environment around the viewer, utilizing the expansive capabilities of VR. The viewer, equipped with a head-mounted display (HMD) specifically the Oculus, experiences the story as if inside it, with narrative vignettes appearing and disappearing in the surrounding darkness. This



Figure 1: Storyboard images for ideation.

project endeavors to transcend traditional 2D storytelling by immersing the viewer in the world of the story, offering a deeper understanding of the dysgraphic experience.

**5.1.1 Process.** The project was developed following a process similar to how a typical, animated short film project would be created. The team started by writing a script for the story, describing the general plot structure, what happens scene-by-scene, as well as cues for the visual and audio components of each scene. The team met with the larger research group each week to give updates on their progress, and several story directions were explored and abandoned based on feedback from the group. Next, visual artists on the team developed concept art to depict the lighting, environment, character, tone, and cinematography for each major scene in the animated experience. Generative AI tools such as MidJourney and Stable Diffusion were used as part of the ideation process to visualize concepts that would be otherwise difficult or time-consuming to render. This approach facilitated quick iteration and more polished presentations of ideas. The final concept art and storyboards were crafted manually through digital painting and drawing.

In the production phase, the platform for this interactive virtual space was Epic Games's Unreal Engine 5, a 3D creation tool with a real-time rendering engine that is commonly used to create games and interactive experiences. Due to time constraints, the original 3D model of the protagonist was replaced with a purchased virtual character, adjusted to align with the concept art. Other assets were created using Maya and Adobe's Substance Painter, with key environmental elements like a sea of crumpled paper balls and floating symbols simulated in SideFX's Houdini for realistic interaction and transformation, enhancing the storytelling aspect of the project.

**5.1.2 Result.** The team reached the end of their project timeline with a working prototype of their immersive narrative that they could view in the VR headset. The project offered a unique blend of firsthand experiences and creative interpretation. This immersive experience helped both participants with ID and other students transcend traditional storytelling methods, while also fostering a deeper understanding of the dysgraphia condition. The project's success lies in its ability to effectively combine personal narratives with immersive animation techniques in VR, creating a powerful and empathetic portrayal of the dysgraphic experience.



Figure 2: Part of the final VR scene of the Through the Eyes.



Figure 3: User's View in AllyChat

## 5.2 AllyChat

Conversational agents hold the potential to significantly enhance the quality of life and independence of individuals with ID by addressing their communication, social interaction, life skills training, and education needs [Federici, 2020]. These agents have proven effective for competence training and well-being management among individuals with ID [De Filippis, 2020]. Moreover, they have been utilized in education, offering personalized tutoring and adapting to students' unique learning styles [Zhang et al., 2018]. This type of tool shows promise for other areas such as job interviewing. In response to the need for inclusive job interview practice for individuals with ID, *AllyChat* was created.

**5.2.1 Process.** A two-part development process was implemented to test chatbot capabilities with the special user group. Given the program students' limited exposure to chatbots, the team collaborated with them to create a chatbot with a Batman persona, a character they knew and felt comfortable chatting with. Usability testing was conducted to gather feedback on the conversational flow between the user and the bot. Generally, the feedback was positive, with students finding the chatbot's personality interesting and engaging. However, they encountered difficulty in generating conversation topics, resulting in shorter interactions.

In the subsequent stages of development, after introducing students to a website chatbot and VR technology, the research team integrated the chatbot into an immersive VR environment. Based on feedback researchers improved the conversational AI agent by

implementing a structured and guided approach. With assistance from the students, the team created a stylized virtual character in an office-lounge area that simulates a semi-public/private space for one-on-one conversation, suitable for both casual conversations with friends and formal job interviews. The stylized model was created by modifying an existing base mesh in Zbrush, featuring idle animations for natural breathing and posture, with Salsa LipSync's real-time audio analysis system approximating mouth movement. *AllyChat* was developed using Unity Engine and OpenXR standard, incorporating the XR Interaction Toolkit for spatial and UI interactions. Unity's HTTP request API facilitated communication between the Oculus VR device and external web services. The audio feature was activated by recording through the Oculus microphone, transcribing to text with Deepgram, forwarding the text to the conversational AI model, and transforming the AI model's response into synthesized speech using AWS Polly for playback through the headset's speakers.

**5.2.2 Result.** The high-fidelity prototype underwent testing by the program students on the Oculus Quest during the regular meeting time. The students were introduced to the application and controls, with no time constraint imposed on the conversation duration. Each student used the application for up to 10 minutes. While the virtual environment appearance received positive feedback, the students found the virtual character's movements to be "kind of weird." Concerns were raised about the speech cadence of the virtual character, specifically regarding speed and length of response. The program director also tested the application and expressed concerns about the language being too advanced for the target audience, feeling that the application was too high level.

## 6 Collaborations Through Co-Design

The core idea of our program was Co-Design, a dynamic and inclusive method of participatory design that united college students and young adults with ID in the transition program. This approach transcended mere collaboration, fostering an integration of shared experiences and ideas. From the beginning of the program, we invited two students from the transition program into our team. Their journey with us was transformative evolving from participants to co-creators after a few weeks of icebreaking sessions through VR game playing. They naturally engaged with the VR design and development teams offering unique insights and enthusiasm. Their impact was profound and multifaceted extending beyond VR technology into the realm of human interactions.

For the neurotypical team members, this collaboration became a journey of personal growth and discovery. We began with our preconceived notions about working with individuals with ID, but through our interactions and shared experiences, these notions were not just challenged but transformed. The process broadened our perspectives, enriching our understanding and empathy. It was about creating a space where everyone could contribute, learn, and grow together, leaving an indelible mark on all participants and reshaping our views on collaboration, innovation, and the power of inclusive design.

## 7 Conclusion

As our project concludes, it is evident that the impact of our work extends far beyond the creation of VR applications. We have not only developed innovative applications but also fostered deeper understandings and acceptances among all participants. The VR experiences 'AllyChat' and 'Through the Eyes,' represent just the beginning of a broader narrative in inclusive technology development.

The true success of this program lies in the newfound confidence and joy observed in our participants from the transition program, as well as the transformative experiences reported by our team members. These outcomes underscore the power of inclusive design practices in fostering personal growth and breaking down societal barriers.

Looking ahead, we envision a future where such collaborative efforts become the norm rather than the exception. We see technology serving as a bridge, connecting people from diverse backgrounds and abilities in a shared journey of discovery and growth. This project has laid the groundwork for more inclusive approaches to technology development, demonstrating how academic institutions can play a pivotal role in creating accessible and empowering digital futures.

Our work highlights the potential for VR and other emerging technologies to not only serve as educational tools but also as catalysts for social change. By continuing to prioritize inclusive design practices and cross-ability collaborations, we can create technologies that truly reflect and serve the diverse needs of all users, ultimately contributing to a more inclusive and understanding society.

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